Part A:

1. **Write a program to convert numbers into words using Enumerations with constructors, methods and instance variables.(INPUT RANGE-0 TO 99999) EX: 36 THIRTY SIX (Console Application)**
    a. click on new project
    b. select java application
    c. write this code inside default class. My default class here is `Main`
    d. write enum outside default class.

```
import java.util.Scanner;

enum NumString {
    ZERO(0), ONE(1), TWO(2), THREE(3), FOUR(4), FIVE(5), SIX(6), SEVEN(7),
EIGHT(8), NINE(9), TEN(10), ELEVEN(11),
    TWELVE(12), THIRTEEN(13), FOURTEEN(14), FIFTEEN(15), SIXTEEN(16),
SEVENTEEN(17), EIGHTEEN(18), NINETEEN(19),
    TWENTY(20), THIRTY(30), FORTY(40), FIFTY(50), SIXTY(60), SEVENTY(70),
EIGHTY(80), NINETY(90);

    int value;

    NumString(int value) {
        this.value = value;
    }

    static String getNumString(int n) {
        for (NumString v : values())
            if (v.value == n)
                return v.toString();
        return "";
    }

    static String convert(int n) {
        if (n <= 20)
            return getNumString(n);

        int rem = n % 10;
        if (rem == 0)
            return getNumString(n);
        else
            return getNumString((n / 10) * 10) + " " + getNumString(rem);
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```java
        System.out.print("Enter a number (0-99999): ");
        int num = sc.nextInt();
        if (num < 0 || num > 99999) {
            System.out.println("Number out of range");
            return;
        }
        if (num == 0) {
            System.out.println(NumString.ZERO.toString());
            return;
        }

        String word = "";

        if (num >= 1000) {
            word += NumString.convert(num / 1000) + " THOUSAND ";
            num %= 1000;
        }

        if (num >= 100) {
            word += NumString.convert(num / 100) + " HUNDRED ";
            num %= 100;
        }

        if (num > 0)
            word += NumString.convert(num);

        System.out.println("Number in words: " + word);
    }
}
```

2. **Find the second maximum and second minimum in a set of numbers using auto boxing and unboxing. (Console Application)**

```java
import java.util.*;

public class Main{
    public static void main(String[] args) {
        TreeSet<Integer> number=new TreeSet<Integer>();
        Scanner sc=new Scanner(System.in);
        int n,secLarge,secSmall,large,small;
        System.out.println("Enter N: ");
        n=sc.nextInt();

        if(n<3){
            System.out.println("N should be greater than 2");
            return;
        }
```

```java
        System.out.printf("Enter %d elements: ",n);
        for(int i=0;i<n;i++)
            number.add(sc.nextInt());

        large=number.last();
        small=number.first();
        secLarge=number.lower(large);
        secSmall=number.higher(small);
        System.out.printf("Second Highest: %d\nSecond Least:
%d\n",secLarge,secSmall);
    }
}
```

3. **Write a menu driven program to create an Arraylist and perform the following operations**
    a. **Adding elements**
    b. **Sorting elements**
    c. **Replace an element with another**
    d. **Removing an element**
    e. **Displaying all the elements**
    f. **Adding an element between two element**

```java
import java.util.*;

public class Main {
    public static void main(String[] args) {
        ArrayList<Integer> list=new ArrayList<Integer>();
        Scanner sc=new Scanner(System.in);
        while(true){
            System.out.println("\n----Menu---\n1.Adding elements\n2.Sorting
elements\n3.Replace an element with another\n4.Removing an
element\n5.Displaying all the elements\n6.Adding an element between two
elements\n7.Exit\n\nEnter your choice: ");
            switch(sc.nextInt()){
                case 1:System.out.println("\nEnter element: ");
                       list.add(sc.nextInt());
                       System.out.println("\nElement added successfully\n");
                       break;
                case 2:Collections.sort(list);
                       System.out.println("List sorted successfully");
                       break;
                case 3:if(list.size()<=0)
```

```
                        System.out.println("\nInsufficient number of 
elements");
                    else{
                        System.out.println("\nEnter old and new element: 
");
                        int oldVal=sc.nextInt();
                        int newVal=sc.nextInt();
                        if(Collections.replaceAll(list, oldVal, newVal))
                            System.out.println("Elements replaced 
successfully");
                        else
                            System.out.printf("%d not found in given 
list\n",oldVal);
                    }
                    break;
            case 4: System.out.println("Enter the element to be removed: 
");
                    Integer element=sc.nextInt(); //should be Integer not 
int otherwise IndexOutOfBoundException
                    if(list.remove(element))
                        System.out.printf("\n%d removed from the 
list\n",element);
                    else
                        System.out.printf("\n%d not found in the 
list\n",element);
                    break;
            case 5: if(list.isEmpty())
                        System.out.println("List is empty\n");
                    else{
                        System.out.println("\nList elements are: \n");
                        Iterator<Integer> itr=list.iterator();
                        while(itr.hasNext())
                            System.out.printf("%d ",itr.next());
                    }
                    break;
            case 6: if(list.size()<2)
                        System.out.println("\nInsuffiencient elements\n");
                    else{
                        System.out.println("\nEnter starting and ending 
element: ");
                        int sidx,eidx,indx;
                        sidx=list.indexOf(sc.nextInt());
                        eidx=list.indexOf(sc.nextInt());
                        if(sidx==-1||eidx==-1)
                            System.out.println("\nEither element not found 
in list\n");
                        else if(eidx-sidx!=1)
```

```
                                System.out.println("Elements should be 
consecutive\n");
                            else{
                                System.out.println("\nEnter new element: ");
                                list.add(eidx, sc.nextInt());
                                System.out.printf("\nElement is added at index 
%d\n",eidx);
                            }
                        }
                        break;
                default: return;
            }
        }
    }
}
```

4. **Write a java program to find words with even number of characters in a string, then swap the pair of characters in those words and also toggle the characters in a given string**

```
import java.util.*;

public class Main{
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the sentence:\n");
        String[] words=sc.nextLine().split("\\s+"); //shpuld be //s it is 
regexp
        String toggle="";
        for(String word: words){
            if(word.length()%2==0){
                String temp="";
                for(int i=0;i<word.length();i+=2)
                    temp=temp+word.charAt(i+1)+word.charAt(i);
                System.out.print(temp+" ");
            }
            for(char c:word.toCharArray()){
                if(Character.isUpperCase(c))
                    toggle+=Character.toLowerCase(c);
                else
                    toggle+=Character.toUpperCase(c);
            }
            toggle+=" ";
        }
        System.out.println("\nToggled sentence: \n"+toggle);
    }
}
```

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5. **Write a Servlet program that accepts the age and name and displays if the user is eligible for voting or not**
    a. **new project->java web**
    b. **inside webpages you will get default index.html.**
    c. **delete entire content and write index.html code inside that file**
    d. **then write click on source package.**
    e. **give the file name as ValidVote.**
    f. **check the check box. very important**
    g. **then configure glassfish server. refer my serverConfig file**

Index.html

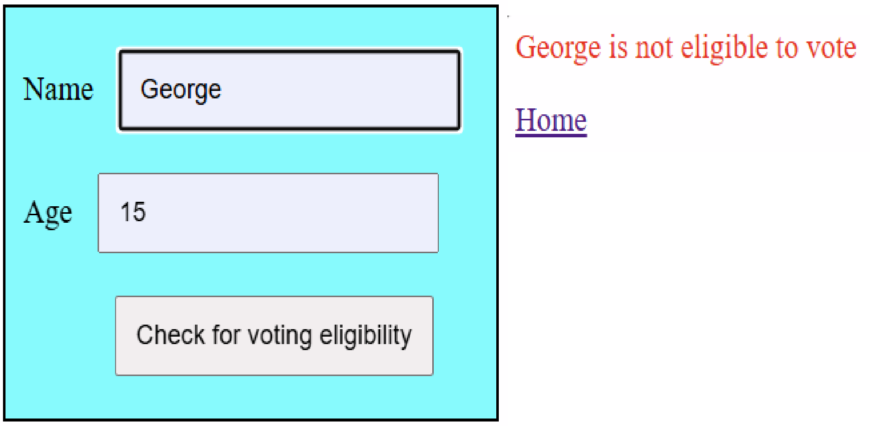
```
<html>
    <head>
        <style>
            form{
                background: cyan;
                border: solid black 2px;
                padding:10px;
                width: fit-content;
                justify-self: center;
            }
            input{
                padding:10px;
                margin:10px;
            }
        </style>
    </head>
    <body>
        <form action="VotingVal">
            Name <input type="text" name="vname"><br>
            Age <input type="text" name="age"><br>
            <input type="submit" value="Check for voting eligibility">
        </form>
    </body>
</html>
```

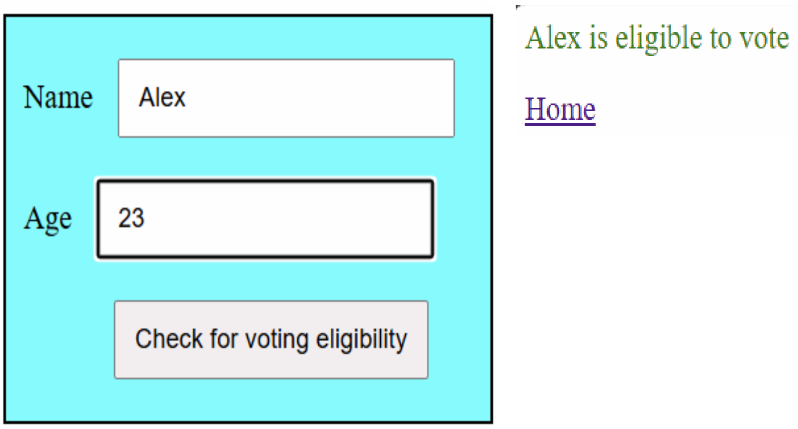
VotingVal.java (Highlighted code is important. other code will be there)

```
public class VotingVal extends HttpServlet {
    //it will contain other code like doGet, doPpst. dont remove them. just
write the code inside try of processRequest method
    protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();
        try {
```

```
                String name=request.getParameter("vname");
//vname is the name of the input field in html form
                int age=Integer.parseInt(request.getParameter("age"));
//age is the name of the input field in html form
                if(age>=18)
                    out.print("<p style=color:green;>"+name+" is eligible to 
vote</p>");
                else
                    out.print("<p style=color:red;>"+name+" is not eligible to 
vote</p>");
                out.print("<a href=index.html>Home</a>");
            }catch(Exception e){
                System.out.println(e);
            }
        }
}
```

output





6. **Write a JSP program to print first 10 Fibonacci and 10 prime numbers.**
    a. **create java web application**
    b. **you will get index.html. delete that file**
    c. **then right click on web files->new->jsp->filename as index**
    d. **you can remove all the code form index.jsp**
    e. **then write following code inside index.jsp**

index.jsp

```jsp
<h1>First 10 Prime numbers:</h1>
<%
    int count = 0, num = 2;
    while (count < 10) {
        boolean isPrime = true;
        for (int i = 2; i <= num / 2; i++)
            if (num % i == 0)
                isPrime = false;

        if (isPrime) {
            out.println(num);
            count++;
        }
        num++;
    }
%>

<h1>First 10 fibonacci Series</h1>
<%
    int first = 0, second = 1, temp;
    for (int i = 0; i < 10; i++) {
        out.println(first);
        temp = first + second;
        first = second;
        second = temp;
    }
%>
```

output

First 10 Prime numbers:

2 3 5 7 11 13 17 19 23 29

First 10 fibonacci Series

0 1 1 2 3 5 8 13 21 34

7. **Write a JSP Program to design a shopping cart to add items, remove item and to display items from the cart using Sessions**
    a. **create 2 jsp file. index.jsp and process.jsp**
    b. **same procedure as before**

index.jsp

```
<h2>Shopping Cart</h2>
<form action="process.jsp" method="post">
    Enter Item Name: <input type="text" name="item"><br>
    Enter price: <input type="number" name='price'><br>
    <input type="submit" name="action" value="Add">
</form>
<br>
<a href="process.jsp">View Cart</a>
```

process.jsp

```
<jsp:useBean id="cart" class="java.util.ArrayList" scope="session" />
<%
    String delItem = request.getParameter("delItem"); //it should be same in
below <a> tag URL
    String iname = request.getParameter("item"); //it should be same in
index.jsp form
    String price = request.getParameter("price");
    if(iname!=null){
        String product="Product Name: "+iname+" Price: "+price;
        cart.add(product);
        response.sendRedirect("process.jsp");
        return;
    }
    if(delItem!=null)
        cart.remove(delItem);
    if(cart.isEmpty())
        out.println("<h3>Cart is empty</h3>");
    else{
        out.println("<h3>Cart Item</h3>");
        for(Object p:cart)
            out.println(p+" <a
href='process.jsp?delItem="+p+"'>Remove</a><br>");
//it should be same as above delItem parameter in URL
            //here dont forget to add '' in href URL because p is string and
it may contain space. if you dont add '' then it will consider space as
separator and it will not work properly.
    }
    out.println("<br><a href='index.jsp'>Continue Shopping</a>");
%>
```

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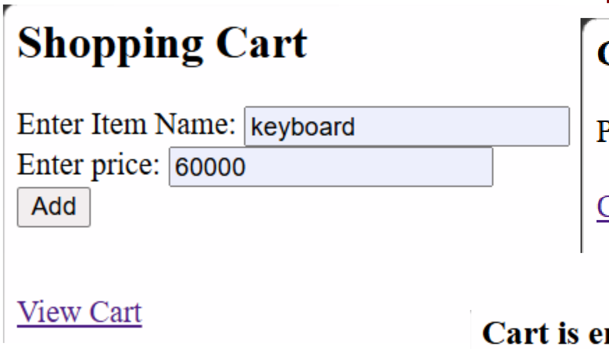
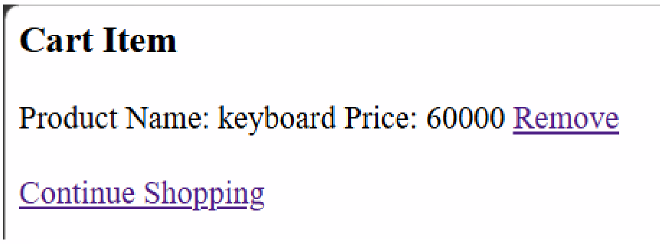
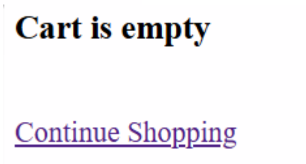
output







8. **Write a java Servlet program to Download a file and display it on the screen**
    a. **here create 3 files**
        i. **index.html it will be by default created.**
        ii. **FileDownload servlet. right click on source package. new->servelet->check the check box**
        iii. **sample.txt. open notepad. write random content. then save this file in the same project file inside WEB folder NOT INSIDE WEBINF.**

index.html

```
<h1>File download</h1>
<a href="FileDownload">Link to Download File</a>
```

FileDownload.java (Servlet) (check only highlighted code)

```
import java.io.IOException;
import java.io.File;
import java.io.OutputStream;
import java.nio.file.Files;


public class FileDownload extends HttpServlet {
    protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        try{
            OutputStream out = response.getOutputStream();
            String path = getServletContext().getRealPath("/sample.txt");
            File f=new File(path);
            if(f.exists()){
                response.setHeader("Content-Disposition", "attachment");
                Files.copy(f.toPath(),out);
```

```
                }
                else
                    response.getWriter().println("File not found!");
        }finally{}

    }
    //keep other code as it is DONT REMOVE THEM
}
```

Part B:

1. **Write a menu driven JDBC program to perform basic operations with Student Table.**
    a. **refer my JDBC Connection document to how to connect java to database**
    b. **create console application**
    c. **add database in service**
    d. **add mysql library. this one is mandatory for all the database project**
    e. **stRegNo is primary key. select it from index dropdown while creating table**

table structure

Field	Type	Null	Key
stRegNo	varchar(10)	NO	PRI
stName	varchar(20)	NO	
stDob	varchar(20)	NO	
stAdrs	varchar(30)	NO	
stClass	varchar(10)	NO	
stCourse	varchar(10)	NO	

```
import java.sql.*;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Connection con;
        PreparedStatement stmt;
        Scanner sc=new Scanner(System.in);
        String name,regno,dob,course,cls,adrs;
        try {
            Class.forName("com.mysql.jdbc.Driver");
            con=DriverManager.getConnection("jdbc:mysql://localhost:3306/javad
b","root",""); //my database name is javadb
            while(true){
                System.out.println("MENU\n1.Add new Student\n2. Delete a
specified students Record\n3. Update Students Address specified students
Record\n4. Search for a particular Student\n5. Exit\nEnter your choice: ");
                switch(sc.nextInt()){
                    case 1:
```

```
                        System.out.println("Enter regno,name,dob,course,class
and adrs: ");
                        regno=sc.next();
                        name=sc.next();
                        dob=sc.next(); //FORMAT: yyyy-mm-dd
                        course=sc.next();
                        cls=sc.next();
                        sc.nextLine(); //important
                        adrs=sc.nextLine();
                        stmt=con.prepareStatement("Insert into student
values(?,?,?,?,?,?)");
                        stmt.setString(1, regno);
                        stmt.setString(2, name);
                        stmt.setString(3, dob);
                        stmt.setString(4, adrs);
                        stmt.setString(5, course);
                        stmt.setString(6, cls);
                        stmt.executeUpdate();
                        System.out.println("Record inserted into table");
                        break;
                    case 2:
                        System.out.println("Enter student id: ");
                        regno=sc.next();
                        stmt=con.prepareStatement("delete from student where
stRegNo=?");
                        stmt.setString(1, regno);
                        if(stmt.executeUpdate()>0)
                            System.out.println("Record deleted successfuly");
                        else
                            System.out.printf("%s ID not found",regno);
                        break;
                    case 3:
                        System.out.println("Enter regno and update adrs: ");
                        regno=sc.next();
                        sc.nextLine();
                        adrs=sc.nextLine();
                        stmt=con.prepareStatement("Update student set stAdrs=?
where stRegNo=?");
                        stmt.setString(1,adrs);
                        stmt.setString(2, regno);
                        if(stmt.executeUpdate()>0)
                            System.out.println("Record updated successfuly");
                        else
                            System.out.printf("%s ID Not found",regno);
                        break;
                    case 4:
                        System.out.println("Enter a regno: ");
                        regno=sc.next();
```

```
                        stmt=con.prepareStatement("select * from student where
stRegNo=?");
                        stmt.setString(1,regno);
                        ResultSet res=stmt.executeQuery();
                        if(res.next())
                            System.out.printf("Student details\nStudent RegNo:
%s\nName: %s\nDOB: %s\nAdrs: %s\nClass: %s\nCourse:
%s",res.getString(1),res.getString(2),res.getString(3),res.getString(4),res.ge
tString(5),res.getString(6));
                        else
                            System.out.printf("%s ID Not found",regno);
                        break;
                    case 5: return;
                }
            }
        }catch (SQLIntegrityConstraintViolationException e1) {
                System.out.println("Reg number already exists");
        }catch (Exception e) {
            System.out.println(e.getMessage());
        }
    }
}
```

output:

```
MENU
1.Add new Student
2. Delete a specified students Record
3. Update Students Address specified students Record
4. Search for a particular Student
5. Exit
Enter your choice:
1
Enter regno,name,dob,course,class and adrs:
101
Alex
13-11-2002
MCA
MCA
2nd main
Record inserted into table
```

```
Enter your choice:
1
Enter regno,name,dob,course,class and adrs:
101
a
a
a
a
a
Reg number already exists
```

```
Enter your choice:
4
Enter a regno:
101
Student details
Student RegNo: 101
Name: Alex
DOB: 13-11-2002
Adrs: 2nd main
Class: MCA
Course: MCAMENU
```

```
Enter your choice:
3
Enter regno and update adrs:
101
3rd main
Record updated successfuly
```

```
Enter your choice:
2
Enter student id:
101
Record deleted successfuly
```

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2. **Write a menu driven JDBC program to perform basic operations with Bank Table.**

```
Table Structure
+-------+-------------+------+-----+
| Field | Type        | Null | Key |
+-------+-------------+------+-----+
| accno | varchar(10) | NO   | PRI |
| name  | varchar(20) | NO   |     |
| adrs  | varchar(20) | NO   |     |
| bal   | float       | NO   |     |
+-------+-------------+------+-----+
```

```
import java.sql.*;
import java.util.Scanner;

public class Main2 {
    public static void main(String[] args) throws ClassNotFoundException,
SQLException {
        Scanner sc = new Scanner(System.in);
        Connection con;
        PreparedStatement stmt;
        ResultSet rs;
        String accno;
        float amt;
        Class.forName("com.mysql.jdbc.Driver");
        con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/javadb", "root", "");
        while (true) {
            try {
                System.out.println("MENU\n 1. Add new Account Holder
information.\n 2. Amount Deposit\n 3. Amount Withdrawal (Maintain minimum
balance 500 Rs)\n 4. Display all information\n 5. Exit\n Enter your choice:
");
                switch (sc.nextInt()) {
                    case 1:
                        System.out.println("Enter Account number, name, adrs
and initial deposite amount(Minimum 500): ");
                        stmt = con.prepareStatement("Insert into bank
values(?,?,?,?)");
                        stmt.setString(1, sc.next());
                        stmt.setString(2, sc.next());
                        sc.nextLine();
                        stmt.setString(3, sc.nextLine());
                        amt = sc.nextFloat();
                        stmt.setFloat(4, amt);
                        if (amt < 500)
                            System.out.println("Minimum amount should be
500");
                        else {
                            stmt.executeUpdate();
```

```
                            System.out.println("Account created
successfully");
                        }
                        break;
                    case 2:
                        System.out.println("Enter account number and amount to
be deposited: ");
                        accno = sc.next();
                        amt = sc.nextFloat();
                        if (amt < 1) {
                            System.out.println("Amount should be positive
number");
                        } else {
                            stmt = con.prepareStatement("update bank set
bal=bal+? where accno=?");
                            stmt.setFloat(1, amt);
                            stmt.setString(2, accno);
                            if (stmt.executeUpdate() > 0)
                                System.out.println("Amount deposited
successfully");
                            else
                                System.out.println("Account not found");
                        }
                        break;
                    case 3:
                        System.out.println("Enter the account number and
amount to be withdrawn: ");
                        accno = sc.next();
                        amt = sc.nextFloat();
                        stmt = con.prepareStatement("select bal from bank
where accno=?");
                        stmt.setString(1, accno);
                        rs = stmt.executeQuery();
                        if (rs.next()) {
                            float tempbal = rs.getFloat(1) - amt;
                            if (tempbal < 500)
                                System.out.println("Maintain minimum
balance...");
                            else {
                                stmt = con.prepareStatement("update bank set
bal=? where accno=?");
                                stmt.setFloat(1, tempbal);
                                stmt.setString(2, accno);
                                stmt.executeUpdate();
                                System.out.println("Amount withdrawn
successfully");
                            }
                        } else
```

```java
                        System.out.println("Account number not found");
                    break;
                case 4:
                    System.out.println("Enter account number: ");
                    accno = sc.next();
                    stmt = con.prepareStatement("Select * from bank where accno=?");
                    stmt.setString(1, accno);
                    rs = stmt.executeQuery();
                    if (rs.next())
                        System.out.printf("------Account information---------\nAccount number: %s\nName: %s\nAddrss: %s\nBalance: %f", rs.getString(1), rs.getString(2), rs.getString(3), rs.getFloat(4));
                    else
                        System.out.println("Account number not found\n");
                    break;
                case 5:
                    return;
                }
            } catch (SQLIntegrityConstraintViolationException e1) {
                System.out.println("Account number already exists");
            } catch (Exception e) {
                System.out.println(e);
            }
        }
    }
}
```

output:

```
MENU
 1. Add new Account Holder information.
 2. Amount Deposit
 3. Amount Withdrawal (Maintain minimum balance 500 Rs)
 4. Display all information
 5. Exit
 Enter your choice:
1
Enter Account number, name, adrs and initial deposite amount(Minimum 500):
101
Goutham
5th main
500
Account created successfully
```

```
 Enter your choice:
1
Enter Account number, name, adrs and initial deposite amount(Minimum 500):
101
a
a
10
Minimum amount should be 500
```

```
 Enter your choice:
1
Enter Account number, name, adrs and initial deposite amount(Minimum 500):
101
a
a
500
Account number already exists
```

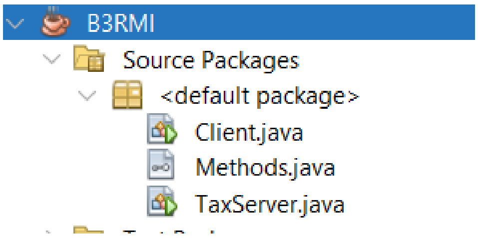
```
 Enter your choice:
4
Enter account number:
101
------Account information---------
Account number: 101
Name: Goutham
Addrss: 5th main
Balance: 500.000000MENU
```

```
 Enter your choice:
3
Enter the account number and amount to be withdrawn:
101
450
Maintain minimum balance...
```

```
 Enter your choice:
3
Enter the account number and amount to be withdrawn:
101
100
Amount withdrawn successfully
```

3. **Write a Java class called Tax with methods for calculating Income Tax. Have this class as a servant and create a server program and register in the rmiregistry. Write a client program to invoke these remote methods of the servant and do the calculations. Accept inputs interactively.**
    a. **create**
        i. **Methods.java (interface)**
        ii. **TaxServer.java (normal java class)**
        iii. **Client.java (normal java class)**
        iv. **create all 3 run inside single package... no need to create separate packages.**

b. **first run Tax.java(Only once)(Right click run)**
c. **then run Client.java (You can run it repeatedly) (Right click run)**



Methods.java

```java
import java.rmi.*;
public interface Methods extends Remote {
    public double incomeTax(int income) throws RemoteException;
}
```

TaxServer.java

```java
import java.rmi.*;
import java.rmi.server.*;
import java.rmi.registry.*;

public class TaxServer extends UnicastRemoteObject implements Methods{
    //generate constructor and implement the method of interface using right click method and bulb method
    public TaxServer() throws RemoteException {}
    public double incomeTax(int income) throws RemoteException {
        if(income<=300000)
            return 0;
        if(income<=600000)
            return income*0.05;
        if(income<=900000)
            return income*0.1;
        if(income<=1200000)
            return income*0.15;
        if(income<=1500000)
            return income*0.2;
        return income*0.3;
    }
    public static void main(String[] args) {
        try{
            LocateRegistry.createRegistry(1099);
            TaxServer s=new TaxServer();
            Naming.rebind("rmi://localhost:1099/serverObj", s);
            //it should be like this
            //if you change the port number... thenm change it here too
            System.out.println("Server is active");
```

```
        }catch(Exception e){
            System.out.println(e);
        }
    }

}
```

Client.java

```
import java.util.Scanner;
import java.rmi.*;
public class Client {
    public static void main(String[] args) {
        try{
            Scanner sc=new Scanner(System.in);
            System.out.println("Enter income: ");
            int income=sc.nextInt();

            Methods
m=(Methods)Naming.lookup("rmi://localhost:1099/serverObj");
            //copy this from server
            double tax=m.incomeTax(income);
            System.out.printf("Income: %d\nTax: %.2f",income,tax);
        }catch(Exception e){
            System.out.println(e);
        }
    }
}
```

Output

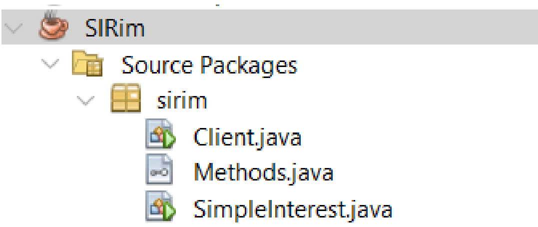
```
Enter income:
45000
Income: 45000
Tax: 0.00
```

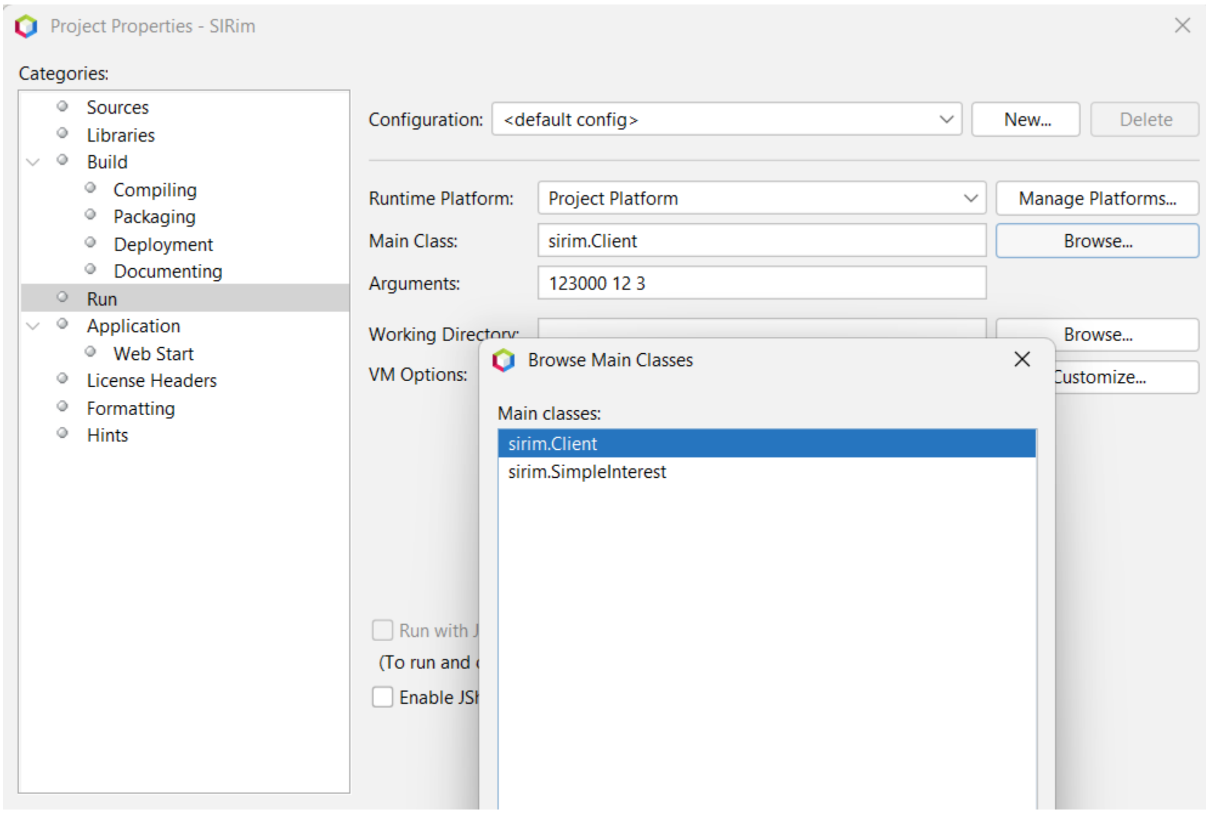
```
Enter income:
1456700
Income: 1456700
Tax: 291340.00
```

```
Enter income:
560000
Income: 560000
Tax: 28000.00
```

4. **Write a Java class called SimpleInterest with methods for calculating simple interest. Have this class as a servant and create a server program and register in the rmiregistry. Write a client program to invoke these remote methods of the servant and do the calculations. Accept inputs at command prompt.**
    a. **create**
        i. **Methods.java (interface)**
        ii. **SimpleInterest.java (server right click run)**

iii. **Client.java (PLAY BUTTON RUN)**
iv. **right click on project->property->run->there select main class as Client.java and set 3 arguments**





Methods.java

```
import java.rmi.*;
public interface Methods extends Remote{
    double calculateSI(double p,double t,double r) throws RemoteException;
}
```

SimpleInterest.java(righr click run)

```
import java.rmi.RemoteException;
import java.rmi.server.*;
import java.rmi.registry.*;
import java.rmi.*;
public class SimpleInterest extends UnicastRemoteObject implements Methods{
```

```java
    //right click generate
    public SimpleInterest() throws RemoteException {}

    public double calculateSI(double p, double t, double r) throws
RemoteException {
        return (p*t*r)/100;
    }

    public static void main(String[] args) {
        try{
            LocateRegistry.createRegistry(1099);
            SimpleInterest s=new SimpleInterest();
            Naming.rebind("rmi://localhost:1099/serveObj", s);
        }catch(Exception e){
            System.out.println(e);
        }
    }
}
```

Client.java

```java
import java.rmi.*;

public class Client {
    public static void main(String[] args) {
        try{
            double p,t,r,res;
            if(args.length!=3){
                System.out.println("Invalid number of args");
                return;
            }
            p=Double.parseDouble(args[0]);
            t=Double.parseDouble(args[1]);
            r=Double.parseDouble(args[2]);

            Methods m=(Methods)
Naming.lookup("rmi://localhost:1099/serveObj");
            res=m.calculateSI(p, t, r);
            System.out.printf("Simple interest %f\n",res);
        }catch(Exception e){
            System.out.println(e);
        }
    }
}
```

5. **Write a Servlet Program to perform Insert, update and View operations on Employee Table**
    a. **create 5 files**
        i. **index.html(default)**
        ii. **EmployeeModel.java(Normal java)**
        iii. **EditEmployee.java(servlet)**
        iv. **ViewEmployee.java(servlet)**
        v. **SaveEmployee.java(servlet)**
    b. **add mysql library**

table structure:

Note: make id col as Primary key, int and AutoIncrement

Field	Type	Null	Key
id	int	NO	PRI
name	varchar(20)	NO	
password	varchar(20)	NO	
email	varchar(30)	NO	
country	varchar(20)	NO	

index.html

```
<h1>Add new employee</h1><br>
        <form action="SaveEmployee">
                Employee Name:  <input type="text" name="name"><br>
                Password:       <input type="password" name="password"><br>
                Email:          <input type="email" name="email"><br>
                Country:        <select name="country">
                    <option value="India">India</option>
                    <option value="Russia">Russia</option>
                    <option value="China">China</option>
                </select><br>
                <input type="submit" value="Save"><br>
                <a href="ViewEmployee">View Employee</a>
        </form>
```

EmployeeModel.java

```
import java.sql.*;
public class EmployeeModel {
    static Connection con;
    static PreparedStatement stmt;
    static{
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
```

```java
            con=DriverManager.getConnection("jdbc:mysql://localhost:3306/javad
b?zeroDateTimeBehavior=CONVERT_TO_NULL","root","");
        } catch (Exception e) {
            System.out.println(e);
        }

    }
    public static int saveEmployee(String name,String password,String
email,String country,String id) throws Exception{
        if(id==null)
            stmt=con.prepareStatement("Insert into employee
values(null,?,?,?,?)");
        else{
            stmt=con.prepareStatement("update employee set
name=?,password=?,email=?,country=? where id=?");
            stmt.setString(5, id);
        }
        stmt.setString(1, name);
        stmt.setString(2, password);
        stmt.setString(3, email);
        stmt.setString(4, country);
        return stmt.executeUpdate();
    }
    public static ResultSet getEmployee(String id) throws Exception{
        if(id==null)
            stmt=con.prepareStatement("select * from employee");
        else{
            stmt=con.prepareStatement("select * from employee where id=?");
            stmt.setString(1,id);
        }
        return stmt.executeQuery();
    }
}
```

SaveEmployee.java(Highlighted code only)

```java
public class SaveEmployee extends HttpServlet {
    protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();
        try{
            String name,password,email,country,id;
            name=request.getParameter("name");
            password=request.getParameter("password");
            email=request.getParameter("email");
            country=request.getParameter("country");
```

```java
            id=request.getParameter("id");
            EmployeeModel.saveEmployee(name, password, email, country,id);
            response.sendRedirect("ViewEmployee");
        }catch(Exception e){
            out.println(e);
        }
    }
    //Dont REMOVE BELOW METHODS
}
```

EditEmployee.java

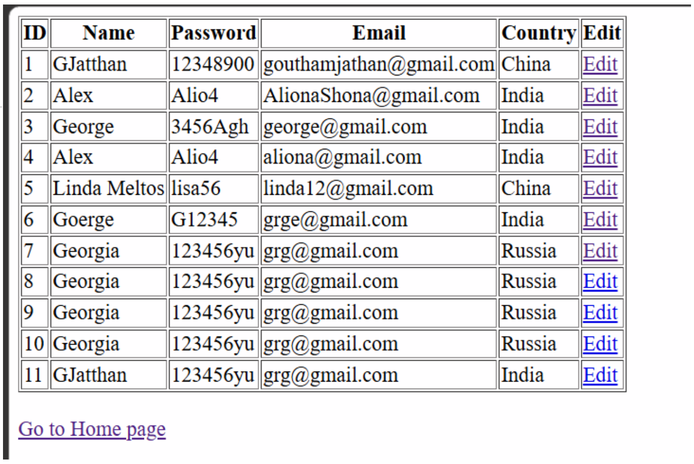
```java
public class EditEmployee extends HttpServlet {
    protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();
        try{
            String id=request.getParameter("id");
            ResultSet row=EmployeeModel.getEmployee(id);
            row.next();
            ArrayList<String> countryList=new
ArrayList<String>(Arrays.asList("India","China","Russia"));
            String country=row.getString(5);
            countryList.remove(country);
            out.println("<H1>Update Employee</h1>"
                    + "<form action=SaveEmployee><pre>"
                    + "<input type=hidden name=id
value="+row.getString(1)+"><br>"
                    + "Name     : <input type=text name=name
value='"+row.getString(2)+"'><br>"
                    + "Password : <input type=password name=password
value="+row.getString(3)+"><br>"
                    + "Email    : <input type=email name=email
value="+row.getString(4)+"><br>"
                    + "Country  : <select name=country>"
                            + "<option value="+country+"
selected>"+country+"</option>"
            );

            for(String c: countryList)
                out.print("<option value="+c+">"+c+"</option>");

            out.println("</select></br>"
                    + "<input type=submit value=Update>"
                    + "</pre></form>");
        }catch(Exception e){
```

```
            out.println(e);
        }
    }
    //Dont REMOVE BELOW METHODS
}
```

ViewEmployee.java

```
public class ViewEmployee extends HttpServlet {
    protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();
        try {
            ResultSet res=EmployeeModel.getEmployee(null);
            out.println("<table border=1px>"
                    + "<tr>"
                    + "<th>ID</th>"
                    + "<th>Name</th>"
                    + "<th>Password</th>"
                    + "<th>Email</th>"
                    + "<th>Country</th>"
                    + "<th>Edit</th>"
                    + "</tr>");
            while(res.next()){
                out.print("<tr>"
                        + "<td>"+res.getInt(1)+"</td>"
                        + "<td>"+res.getString(2)+"</td>"
                        + "<td>"+res.getString(3)+"</td>"
                        + "<td>"+res.getString(4)+"</td>"
                        + "<td>"+res.getString(5)+"</td>"
                        + "<td><a
href=EditEmployee?id="+res.getInt(1)+">Edit</a></td>"
                        + "</tr>");
            }
            out.print("</table><br>"
                    + "<a href=index.html>Go to Home page</a>");
        }catch(Exception e){
            out.println(e);
        }
    }
    //Dont REMOVE BELOW METHODS
}
```

output



ID	Name	Password	Email	Country	Edit
1	GJatthan	12348900	gouthamjathan@gmail.com	China	Edit
2	Alex	Alio4	AlionaShona@gmail.com	India	Edit
3	George	3456Agh	george@gmail.com	India	Edit
4	Alex	Alio4	aliona@gmail.com	India	Edit
5	Linda Meltos	lisa56	linda12@gmail.com	China	Edit
6	Goerge	G12345	grge@gmail.com	India	Edit
7	Georgia	123456yu	grg@gmail.com	Russia	Edit
8	Georgia	123456yu	grg@gmail.com	Russia	Edit
9	Georgia	123456yu	grg@gmail.com	Russia	Edit
10	Georgia	123456yu	grg@gmail.com	Russia	Edit
11	GJatthan	123456yu	grg@gmail.com	India	Edit

Go to Home page

6. **Write a java JSP program to get student information through a HTML and create a JAVA Bean Class, populate Bean and Display the same information through another JSP**
   a. **create**
      i. **index.html (default)**
      ii. **process.jsp**
      iii. **display.jsp**
      iv. **StudentBean.java(in source package, normal java class)**

StudentBean.java (Generate ctor, getter, setter using right click method)

Note: create this class inside models package.. IMPORTANT

```
public class StudentBean implements Serializable{
    private String name;
    private int rollNo,age;

    public StudentBean() {
    }

    public String getName() {
        return name;
    }

    public int getRollNo() {
        return rollNo;
    }
```

```java
    public int getAge() {
        return age;
    }

    public void setName(String name) {
        this.name = name;
    }

    public void setRollNo(int rollNo) {
        this.rollNo = rollNo;
    }

    public void setAge(int age) {
        this.age = age;
    }
}
```

index.html

Note: textbox name should be same as studentBean variable name

```html
<h2>Enter Student Details</h2>
        <form action="process.jsp" method="post">
            Name: <input type="text" name="name"><br><br>
            Roll No: <input type="text" name="roll"><br><br>
            Age: <input type="text" name="age"><br><br>
            <input type="submit" value="Submit">
        </form>
```

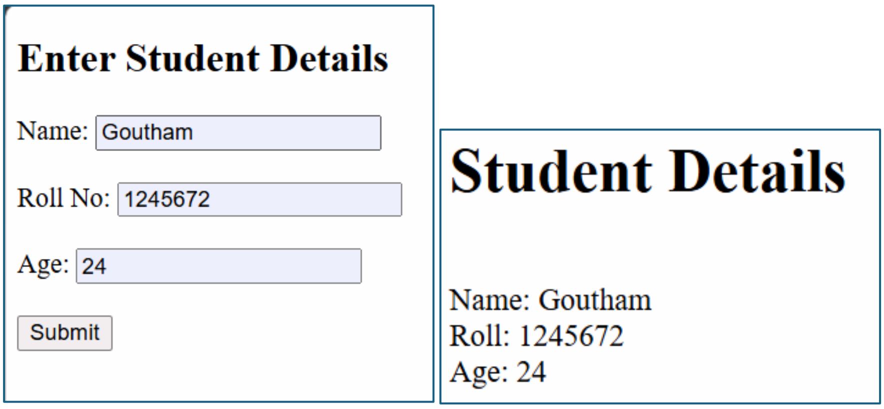
process.jsp

```jsp
<%@ page import="models.StudentBean" contentType="text/html"
pageEncoding="UTF-8"%>
<jsp:useBean id="student" class="models.StudentBean" />
<jsp:setProperty name="student" property="*"/>
<jsp:forward page="display.jsp" />
```

display.jsp

```jsp
<jsp:useBean id="student" class="models.StudentBean" scope="request" />
<h2>Student Information</h2>
Name: <jsp:getProperty name="student" property="name" /><br><br>
Roll No: <jsp:getProperty name="student" property="roll" /><br><br>
Age: <jsp:getProperty name="student" property="age" /><br><br>
```

Output



7. **Write a menu driven program to create a linked list and perform the following operations.**
   a. **to Insert some Elements at the Specified Position**
   b. **swap two elements in a linked list**
   c. **to Iterate a LinkedList in Reverse Order**
   d. **to Compare Two LinkedList**
   e. **to Convert a LinkedList to ArrayList**

```
public class Main {
    public static void main(String[] args) {
        LinkedList<Integer> list=new LinkedList<Integer>();
        Scanner sc=new Scanner(System.in);
        while(true){
            System.out.println("\nMENU\n1.Insert some Elements at the Specified Position\n2.Swap two elements in a linked list\n3.Iterate a LinkedList in Reverse Order\n4.Compare Two LinkedList\n5.Convert a LinkedList to ArrayList\n6.Exit\n\nEnter your choice: ");
            switch(sc.nextInt()){
                case 1: System.out.println("Enter the element to be inserted: ");
                        int ele=sc.nextInt();
                        System.out.printf("Enter the position from 0 to %d: ",list.size());
                        int indx=sc.nextInt();
                        if(indx<0||indx>list.size())
                            System.out.println("Invalid index");
                        else{
                            list.add(indx, ele);
                            System.out.printf("List after insertion: %s",list);
                        }
                        break;
                case 2: if(list.size()<2)
                            System.out.println("Insuficient elements\n");
                        else{
```

```
                            System.out.println("Enter first and second 
element: ");
                            int indx1=list.indexOf(sc.nextInt());
                            int indx2=list.indexOf(sc.nextInt());
                            if(indx1==-1||indx2==-1)
                                System.out.println("Element not found to 
swap\n");
                            else{
                                Collections.swap(list,indx1,indx2);
                                System.out.printf("List after swapping: 
%s",list);
                            }
                        }
                        break;
                case 3: if(list.size()<=0)
                            System.out.println("List is empty");
                        else{
                            System.out.println("List in reverse order: \n");
                            Iterator itr=list.descendingIterator();
                            while(itr.hasNext())
                                System.out.printf("%d ",itr.next());
                        }
                        break;
                case 4: LinkedList<Integer> temp=new LinkedList<Integer>();
                        sc.nextLine();
                        System.out.println("Enter new list elements separated 
by space: ");
                        Scanner nsc=new Scanner(sc.nextLine());
                        while (nsc.hasNextInt())
                            temp.add(nsc.nextInt());
                        System.out.printf("Contents of original list: %s\nTemp 
list: %s\n",list,temp);
                        if(list.equals(temp))
                            System.out.println("Both the list are same ");
                        else
                            System.out.println("linked lists are not same");

                        break;
                case 5: if(list.size()<=0)
                            System.out.println("List is empty");
                        else{
                            ArrayList<Integer> alist=new 
ArrayList<Integer>(list);
                            System.out.printf("Converted array list: 
%s",alist);
                        }
                        break;
                case 6: return;
```

```
            }
        }

    }
}
```

output:

```
MENU
1.Insert some Elements at the Specified Position
2.Swap two elements in a linked list
3.Iterate a LinkedList in Reverse Order
4.Compare Two LinkedList
5.Convert a LinkedList to ArrayList
6.Exit

Enter your choice:
1
Enter the element to be inserted:
30
Enter the position from 0 to 0: 0
List after insertion: [30]
```

```
Enter your choice:
1
Enter the element to be inserted:
50
Enter the position from 0 to 1: 1
List after insertion: [30, 50]
```

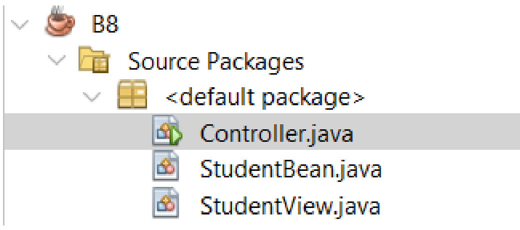
```
Enter your choice:
3
List in reverse order:

50 78 30
```

```
Enter your choice:
2
Enter first and second element:
50 78
List after swapping: [30, 50, 78]
```

```
Enter your choice:
5
Converted array list: [30, 50, 78]
```

```
Enter your choice:
4
Enter new list elements separated by space:
30 50 78
Contents of original list: [30, 50, 78]
Temp list: [30, 50, 78]
Both the list are same
```

8. **Implement a java application based on the MVC design pattern. Input student Rolnlo, name ,marks in three subject calculate result and grade and display the result in neat format.**
    a. **create 3 files in default package. NO NEED TO CREATE SEPARATE PACKAGES**
        i. **StudentBean.java**
        ii. **StundetView.java**
        iii. **Controller.java (Main class)**



StudentBean.java (Generate it using getter setters) (Highlighted code important)

```
public class StudentBean {
    int rno,m1,m2,m3;
    String name,grade;
    float percent;

    public StudentBean(int rno, String name, int m1, int m2, int m3) {
        this.rno = rno;
        this.m1 = m1;
        this.m2 = m2;
        this.m3 = m3;
        this.name = name;
    }

    public int getRno() {
        return rno;
    }

    public int getM1() {
        return m1;
    }

    public int getM2() {
        return m2;
    }

    public int getM3() {
        return m3;
    }

    public String getName() {
        return name;
    }
```

```java
    public String getGrade() {
        return grade;
    }

    public float getPercent() {
        return percent;
    }

    public void calculateResult(){
        percent=(m1+m2+m3)/3;
        if (percent > 90)
            grade = "A";
        else if (percent >= 80)
            grade = "B";
        else if (percent >= 70)
            grade = "C";
        else if (percent >= 60)
            grade = "D";
        else
            grade = "E";
    }
}
```

StudentView.java

```java
public class StudentView {
    public StudentView(StudentBean s) {
        System.out.printf("Student Name:%s\nRollNumber: %d\nM1: %d\nM2: 
%d\nM3: %d\nPercentage: %f\nGrade: 
%s",s.getName(),s.getRno(),s.getM1(),s.getM2(),s.getM3(),s.getPercent(),s.getG
rade());
    }
}
```

Controller.java

```java
public class Controller {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter Roll No, Name and Marks in 3 subject: ");
        int roll = sc.nextInt();
        String name = sc.next();
        int m1 = sc.nextInt();
        int m2 = sc.nextInt();
        int m3 = sc.nextInt();
        StudentBean student = new StudentBean(roll,name,m1,m2,m3);
        student.calculateResult();
        StudentView view = new StudentView(student);
```

```
    }
}
```

Output

```
Enter Roll No, Name and Marks in 3 subject:
101
Alex
90
78
98
Student Name:Alex
RollNumber: 101
M1: 90
M2: 78
M3: 98
Percentage: 88.000000
Grade: B
```